Delhi Public School

Subject: Physics | Grade: 11th | Class: 11th

Time: 2 Hours Maximum Marks: 16

Name: ________________________ Roll No: ___________ Section: ___________

General Instructions:

1. *Read all questions carefully before answering.*
2. *Write neatly and legibly.*

Section A

This section contains 5 multiple-choice questions. Each question carries 1 mark. Select the single correct option. Time Allowed: 1 hour 30 minutes (Total Exam).

Q1. The dimensional formula of Universal Gravitational Constant (G) is: [Easy] **[1]**

a. [M^-1 L^3 T^-2]
b. [M L^3 T^-2]
c. [M^-1 L^2 T^-2]
d. [M^-2 L^3 T^-1]

Q2. A vehicle travels the first half of distance L with speed v1 and the remaining half distance with speed v2. The average speed of the vehicle is: [Easy] **[1]**

a. (v1 + v2) / 2
b. (2 * v1 * v2) / (v1 + v2)
c. (v1 * v2) / (v1 + v2)
d. sqrt(v1 * v2)

Q3. At what angle of projection with the horizontal will the maximum height of a projectile be equal to its horizontal range? [Moderate] **[1]**

a. theta = tan^-1(1)
b. theta = tan^-1(2)
c. theta = tan^-1(4)
d. theta = 45 degrees

Q4. The work done by a conservative force along a closed path/loop is always: [Moderate] **[1]**

a. positive
b. negative
c. zero
d. dependent on the nature of the force

Q5. If the kinetic energy of a body is increased by 300%, what is the percentage increase in its momentum? [Hard] **[1]**

a. 50%
b. 100%
c. 150%
d. 200%

Section B

This section contains 3 short-answer questions. Each question carries 2 marks. Write precise answers.

Q6. Explain why rain drops fall with a constant terminal velocity instead of accelerating continuously under gravity. [Easy] **[2]**

Q7. A block of mass 2 kg is placed on a horizontal surface. If the coefficient of static friction between the block and the surface is 0.4, find the frictional force acting on the block when a horizontal force of 5 N is applied to it. (Take g = 10 m/s^2) [Moderate] **[2]**

Q8. State Kepler's second law of planetary motion (Law of Areas) and show that it is a direct consequence of the conservation of angular momentum. [Hard] **[2]**

Section C

This section contains 5 long-answer questions. Each question carries 1 mark. Write structured derivations and explanations as required.

Q9. Derive the mathematical expression for the variation of acceleration due to gravity (g) with depth (d) below the surface of the Earth. [Moderate] **[1]**

Q10. State and prove Bernoulli's principle for the flow of an ideal, non-viscous, incompressible fluid in a streamlined flow. [Moderate] **[1]**

Q11. State the law of equipartition of energy. Using this law, calculate the molar specific heat capacities at constant volume (Cv) and constant pressure (Cp) for a rigid diatomic gas. [Easy] **[1]**

Q12. What is a simple pendulum? Derive an expression for its time period and show that for small angular displacements, its motion is Simple Harmonic Motion (SHM). [Moderate] **[1]**

Q13. Discuss the formation of standing waves in a stretched string of length L fixed at both ends. Obtain the general mathematical equation for standing waves and derive expressions for the frequencies of the first three harmonics. [Hard] **[1]**

— End of Question Paper —